single-stranded, positive-sense (+) **RNA genome** (Figure 10.10). A positive-sense RNA genome can be directly translated into **viral proteins**, the RNA genome encodes **seven nonstructural proteins** and **three structural proteins**. One of the structural proteins forms the envelope. The RNA genome forms a nucleocapsid along with copies of the 12-kDa capsid protein.

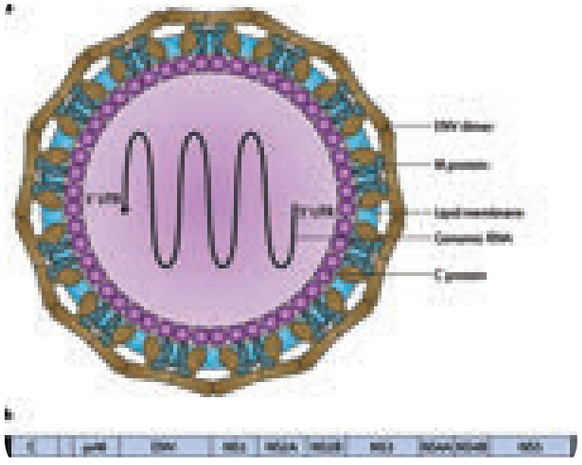


Figure 10.10: Structure of Zika Virus

Viral genome replication depends on the making of double-stranded RNA from the single-stranded, positive-sense RNA (ssRNA(+)) genome followed by **transcription** and **replication** to provide viral mRNAs and **new ssRNA(+) genomes**.

Pathogenesis and Clinical features

Zika virus replicates in the mosquito's **mid gut epithelial cells** and then its salivary gland cells. After 5–10 days, the virus can be found in the mosquito's saliva. If the mosquito's saliva is inoculated into **human skin**, the virus can infect epidermal keratinocytes, skin fibroblasts in the skin and the Langerhans cells. The pathogenesis of the virus is hypothesized to continue with a **spread to lymph nodes** and the **bloodstream**.

Zika virus is primarily transmitted by the **bite of an infected mosquito** from the *Aedes* genus, mainly *Aedes aegypti*. The mosquitoes usually bite during the day, peaking during early morning and late afternoon or evening. This is the same mosquito that transmits dengue, chikungunya and yellow fever. Zika virus is also transmitted from mother to fetus during pregnancy, through sexual contact, transfusion of blood and blood products, and organ transplantation.

The incubation period of Zika virus disease is estimated to be **3–14 days**. The majority of people infected with Zika virus **do not develop symptoms**. Symptoms are generally mild including fever, rash, conjunctivitis, muscle and joint pain, malaise, and headache, and usually last for 2–7 days. **Zika fever** (also known as Zika virus disease) is an illness caused by the Zika virus. Zika virus infection during pregnancy is a cause of **microcephaly** and other **congenital abnormalities** in the developing fetus and newborn. Zika infection in pregnancy also results in pregnancy complications such as fetal loss, stillbirth, and preterm birth.

Laboratory diagnosis

Virus can be demonstrated from the blood or other body fluids, such as urine or semen.

Zika virus grow well in a variety of mammalian and insect cell lines. Zika virus is identified by **NAAT–** Nucleic acid Amplification test, Zika Antigen is detected by ELISA and PCR. **Zika Antibody** IgM is detected by MAC - ELISA, IgG by ELISA and by **PRNT**-plaque reduction neutralization test.

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